Combined Control of Power Electronics and Linear Actuator for Inverse Kinematics Applications

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I. Introduction

The Michigan Mars Rover Team has recently developed the need for inverse kinematics on their robotic arm to perform an autonomous typing task successfully. The development of the inverse kinematics has gone smoothly except for the reoccurring problem of notable differences between the commanded and observed joint speeds (e.g. if joint B is set to 40 rpm, and joint C is set to 50 rpm, then if joint B is actually 30 rpm, the robot end-effector cannot reach its target successfully).

It is hypothesized that these differences arise from poorly modeled dynamics and neglected startup dynamics for each joint. Current modeling assumes second order differential equation type behavior for the buck that feeds the bus voltage, the h-bridge, and actuator with its motor and gearbox. With this setup, only PID was needed for sufficient control. However, if these dynamics can be better modeled and controlled, precise tracking control of each joint should be possible. For this study, only the linear actuator is considered.

II. Approach

A. System Identification

The first task in any controls problem is system identification. The proposed system includes a buck, h-bridge, a motor, and a gearbox. We will model the load torque, but we will just treat it as an extra linear term without considering its effect on inertia just yet. The important constants that are not free to choose are listed below:

- Input voltage V_{in}
- Motor inductance L_A
- Motor winding resistance R_A
- Motor/gearbox kinetic friction coefficient τ_c
- Motor/gearbox viscous friction coefficient B
- Motor torque constant K_t
- Motor back-emf constant K_e
- Motor inertia J

Input voltage is very easily determined as it is battery voltage which has already been specified by the team requirements: $V_{in} \in [34, 42]V$ with nominal $V_{in} = 36V$ Motor winding inductance and resistance were both determined by use of an LCR meter. Because the impedance of the motor changes significantly with injected frequency and current due to nonlinear effects like the skin effect, eddy currents, and saturation, the motor resistance and inductance were measured at a all of the possible frequency settings. The settings may not be accurate enough, however, because Mrover uses a fairly high pwm frequency to drive the motor. Below are the measurements:

TABLE I
LCR Measurements

Test Frequency (Hz)	Motor Inductance (mH)	Motor Resistance (Ω)
100	3.260	3.436
120	3.204	3.539
1000	2.192	7.184
10000	1.366	33.65

The motor torque constant can be determined from an equation for the force transfer mechanics of a linear actuator and values provided by the manufacturer. The motor torque constant is:

$$K_t = \frac{K_F p}{2\pi\eta N}$$

where K_F is the force constant, p is the pitch lead of the screw, η is the screw efficiency, and N is the gear ratio. K_F can be calculated from the following equation:

$$K_F = \frac{F}{I}$$

where I is the current at load force F. This data is provided by the manufacturer. Plugging in all of the manufacturer constants in and estimating $\eta \approx 0.3$ (not provided by manufacturer), K_t is calculated as:

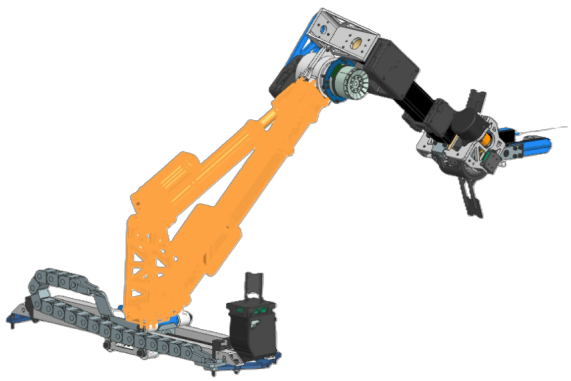
$$K_t = \frac{\frac{890}{3.1} \times 0.00737}{2\pi \times 0.3 \times 35} = 0.032\frac{Nm}{A}$$

The motor back-emf constant can be determined from an equation for the electrical characteristic of the motor winding and values provided by the manufacturer.

The motor back-emf constant is:

$$K_e = \frac{V - Ri}{\omega}$$

where V is the applied voltage to the motor, R is the DC resistance of the motor windings, i is the motor steady state



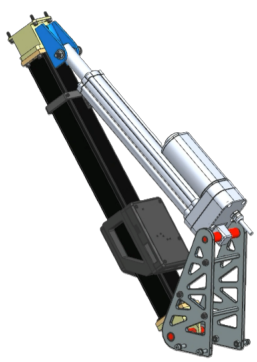


Fig. 1. MRover Robotic Arm

TABLE II
FINAL PLANT PARAMETER TABLE

Parameter	Unit
$0.04 < K_e < 0.044$	$\frac{Vs}{rad}$
$0.032 < K_t < 0.04$	$\frac{Nm}{A}$
$3 < R_a < 7$	Ω
$2 < La < 3.4$	mH
$0 < J_m < 6 \times 10^{-7}$	$kg * m^2$
$0 < B_m < 4 \times 10^{-5}$	$\frac{N*s}{rad}$
$100 < C < 500$	μF
$100 < L < 300$	μH
$f_{sw} = 100$	kHz
$f_{ctrl} = 5$	kHz

current, and ω is the angular speed of the motor. When substituting approximate values from hardware tests for voltage, resistance, and current and the no-load angular speed from the manufacturer:

$$K_e = \frac{24 - 3.3 \times 0.8}{419} = 0.0435 \frac{Vs}{rad}$$

Motor inertia J was estimated to be 6×10^{-7} $kg\ m^2$ from a reference motor with similar size.

Motor viscous friction B was estimated to be 4×10^{-5} by comparing model current values with tested real current values.

B. State Space Modeling

The four circuit configurations are given as:

TABLE III
FOUR CIRCUIT CONFIGURATIONS

u_1	u_2
0	0
0	1
1	0
1	1

For the first configuration, the derived equations are:

$$\begin{aligned} L\frac{di_L}{dt} &= -V_C \\ C\frac{dV_C}{dt} &= i_L - \frac{V_C}{R} + i_{L_A} \\ L_A\frac{di_{L_A}}{dt} &= -V_{R_A} - V_C - V_\tau - K_e\omega \\ J\frac{d\omega}{dt} &= K_t i_{L_A} - B\omega - \tau_c - \tau_L \end{aligned}$$

The second configuration:

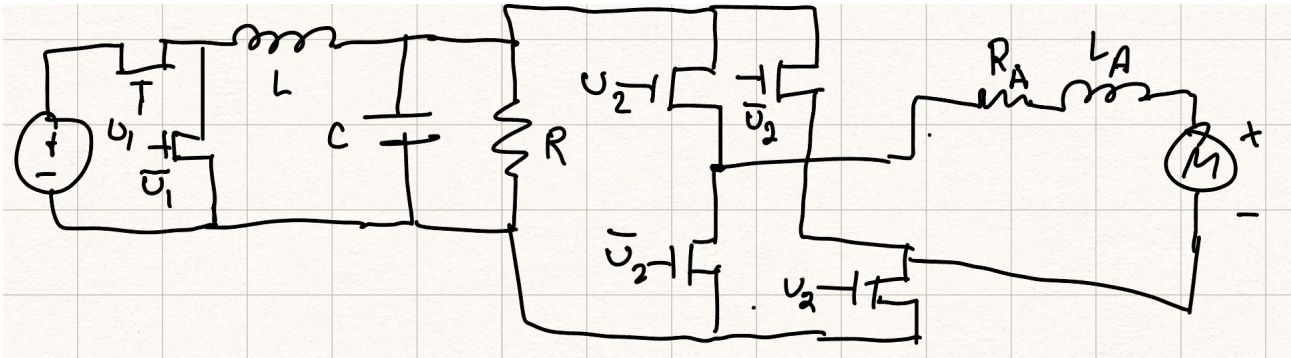


Fig. 2. Circuit Diagram Labeling Switches for State Space Derivation

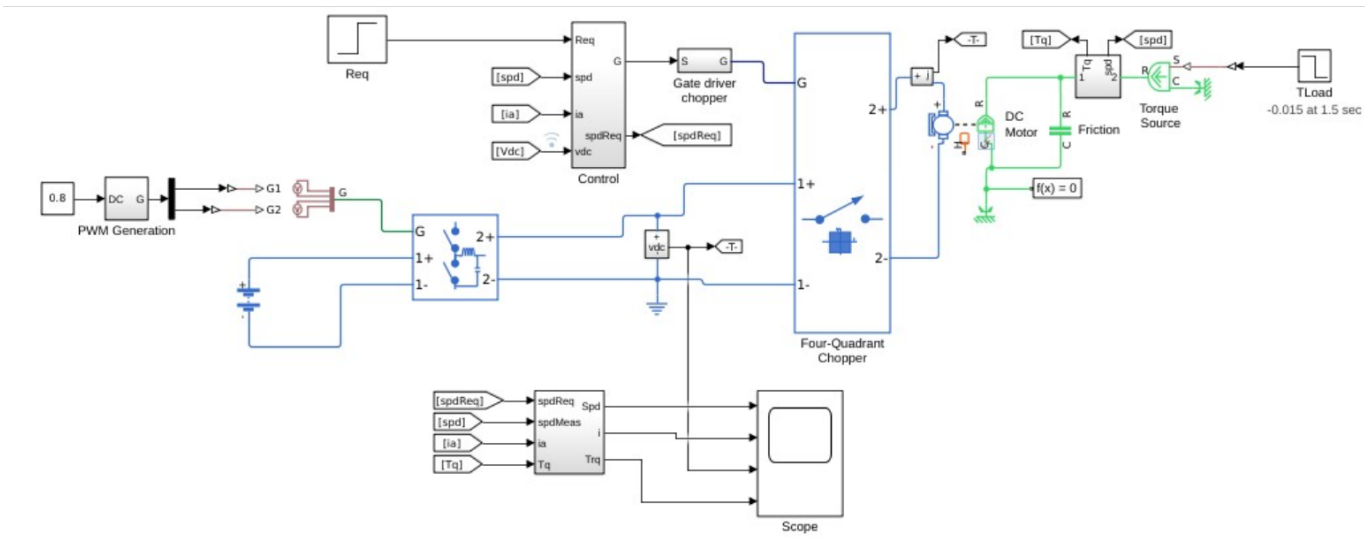


Fig. 3. Initial Simulink Plant model based on MATLAB example

$$
\begin{aligned}
L\frac{di_L}{dt} &= -V_C \\
C\frac{dV_C}{dt} &= i_L - \frac{V_C}{R} + i_{L_A} \\
L_A\frac{di_{L_A}}{dt} &= V_C - V_{R_A} - V_\tau - K_e\omega \\
J\frac{d\omega}{dt} &= K_t i_{L_A} - B\omega - \tau_c - \tau_L
\end{aligned}
$$

The third configuration:

$$
\begin{aligned}
L\frac{di_L}{dt} &= V_{in} - V_C \\
C\frac{dV_C}{dt} &= i_L - \frac{V_C}{R} + i_{L_A} \\
L_A\frac{di_{L_A}}{dt} &= -V_C - V_{R_A} - V_\tau - K_e\omega \\
J\frac{d\omega}{dt} &= K_t i_{L_A} - B\omega - \tau_c - \tau_L
\end{aligned}
$$

The fourth configuration:

$$
\begin{aligned}
L\frac{di_L}{dt} &= V_{in} - V_C \\
C\frac{dV_C}{dt} &= i_L - \frac{V_C}{R} - i_{L_A} \\
L_A\frac{di_{L_A}}{dt} &= V_C - V_{R_A} - V_\tau - K_e\omega \\
J\frac{d\omega}{dt} &= K_t i_{L_A} - B\omega - \tau_c - \tau_L
\end{aligned}
$$

When all equations are combined and put in bilinear form:

$$
\begin{bmatrix} \dot{i}_L \\ \dot{V}_C \\ \dot{i}_{La} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} & 0 & 0 \\ \frac{1}{C} & 0 & -\frac{1}{C} & 0 \\ 0 & -\frac{1}{L_a} & -\frac{R_a}{L_a} & -\frac{K_e}{L_a} \\ 0 & 0 & \frac{K_t}{J_m} & -\frac{B_m}{J_m} \end{bmatrix} \begin{bmatrix} i_L \\ V_C \\ i_{La} \\ \omega \end{bmatrix}
+ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{C} & 0 \\ 0 & \frac{2}{L_A} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_L \\ V_C \\ i_{La} \\ \omega \end{bmatrix} u_2 + \begin{bmatrix} \frac{V_{in}}{L} \\ 0 \\ 0 \\ 0 \end{bmatrix} u_1
$$

Clearly there are nonlinear terms that rely on u_2 so a linear control method cannot be used without averaging and a small signal model.

III. Average Modeling

We determined that treating the buck converter as an averaged model with a duty cycle $d \in [0, 1]$ would hold as the synchronous buck converter frequency is much higher than the RL time constant of the motor or the switching constant of the drive. This does not fully decouple the systems but leaves a more realistic control landscape as cycle-by-cycle DC-DC control is usually only achieved by ASICs.

The plant dynamics become:

$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_C \\ \dot{i}_{La} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} & 0 & 0 \\ \frac{1}{C} & 0 & -\frac{1}{C} & 0 \\ 0 & -\frac{1}{L_a} & -\frac{R_a}{L_a} & -\frac{K_e}{L_a} \\ 0 & 0 & \frac{K_t}{J_m} & -\frac{B_m}{J_m} \end{bmatrix} \begin{bmatrix} i_L \\ V_C \\ i_{La} \\ \omega \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{C} & 0 \\ 0 & \frac{2}{L_A} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_L \\ V_C \\ i_{La} \\ \omega \end{bmatrix} u_2 + \begin{bmatrix} \frac{V_{in}}{L} \\ 0 \\ 0 \\ 0 \end{bmatrix} d$$

IV. Control Policies/Objectives

As the actuator is part of a robotic system, safety and crash-tolerance are important concerns. It should be able to crash without burning out the motor or any of its components. Consequently we are adding a control policy to keep $|i_{La}| < 5A$

In addition, control lyapunov functions can be applied to the system as it takes the form:

$$\dot{x} = f(x) + g(x)u$$

Where

$$f(x) = \begin{bmatrix} \frac{-V_C}{L} \\ \frac{i_L - i_{La}}{C} \\ \frac{-V_C - R_a i_{La} - K_e \omega}{L_a} \\ \frac{K_t i_{La} - B_m \omega}{J_m} \end{bmatrix} g(x) = \begin{bmatrix} \frac{V_{in}}{L} & 0 \\ 0 & \frac{2 i_{La}}{C} \\ 0 & \frac{2 V_C}{L_a} \\ 0 & 0 \end{bmatrix} u = \begin{bmatrix} d \\ u_2 \end{bmatrix}$$

The energy Lyapunov equation for the system is:

$$V(x) = \frac{x_1^2}{2} + \frac{x_2^2}{2} + \frac{x_3^2}{2} + \frac{x_4^2}{2}$$

Where

$$x_1 = i_L, x_2 = V_c, x_3 = i_{La}, x_4 = \omega$$

$$\dot{V}(x) = x_1 \dot{x_1} + x_2 \dot{x_2} + x_3 \dot{x_3} + x_4 \dot{x_4}$$

By varying the control variables d and u_2 we can set requirements for lyapunov stability. (done by forcing $\dot{x_n} = -kx_n$ for each state variable) The system is stable i.s.L when $d = \frac{-hLi_L + V_c}{V_{in}}, h > 0$ and for such times as $\exists k > 0, m > 0$ $u_2 = \frac{-mLV_c - i_L}{2i_L a} = \frac{-kL_a i_L a + K_e \omega}{2V_c}$

V. Timescale Separation

Because the system contains both electrical and mechanical components, there are time constants associated with each, with the electrical being much faster than the mechanical. Nonlinear controllers were implemented for each timescale.

Model Predictive Control (MPC) was chosen for the outer loop because of its real-time capabilities and tunability. The cost function can be carefully controlled to provide the desired behaviors. MPC enabled angular speed trajectory tracking by creating bus voltage (V_C) and motor current (i_{LA}) setpoints.

Sliding Mode Control (SMC) was chosen for the inner loop of the controller due to its high-speed control capabilities and robustness. One SMC controller was used for the buck circuit feeding the H-bridge, taking in the V_C setpoint from the MPC controller and outputting the switching function to S_1. The other SMC controller was used for the H-bridge circuit feeding the motor, taking in the $I_L A$ setpoint from the MPC controller and outputting the switching function S_2.

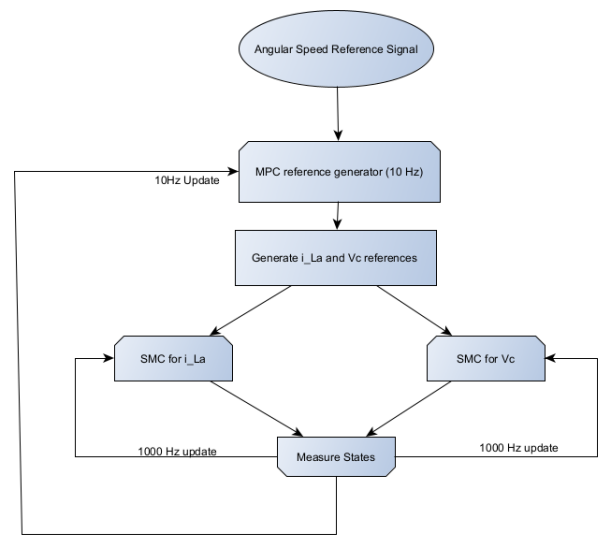


Fig. 4. Flowchart detailing control architecture

VI. Sliding Mode Controller Derivations

A. Switching Function S_1

The buck requires a second-order sliding surface because the bus voltage does not directly contain the control input u_1. Instead the buck inductor current i_L contains the d dependence and the surface must be a two-state system. u_1 is averaged and denoted as the duty cycle d.

The buck inductor current and output capacitor voltage equations are given below:

$$\frac{di_L}{dt} = \frac{1}{L}(-V_C + V_{in} d)$$
$$\frac{dV_C}{dt} = \frac{1}{C} i_L - \frac{1}{RC} V_C - \frac{1}{C} i_{LA}$$

The error and error rate are defined as:

$$e_v = V_{targ} - V_C$$
$$\dot{e_v} = -\dot{V_C}$$

The sliding surface and its derivative are defined as:

$$\begin{aligned}
S_1 &= \dot{e_v} + a_0 e_v \\
&= -\frac{1}{C} i_L + \frac{1}{RC} V_C + \frac{1}{C} i_{LA} + a_0 (V_{targ} - V_C) \\
\dot{S_1} &= -\frac{1}{C}[-\frac{1}{L} V_C + \frac{V_{in}}{L} d] \\
&\quad + (\frac{1}{RC} - a_0)[\frac{1}{C} i_L - \frac{1}{RC} V_C - \frac{1}{C} i_{LA}] \\
&\quad + \frac{1}{C} \dot{i_{LA}}
\end{aligned}$$

From Lyapunov stability theory, $S\dot{S} < 0$ must be true in order to guarantee the error will go to 0. For convenience, the $\dot{S}$ is redefined in terms of functions $h(x)$ and $g(x)$:

$$\dot{S} = h(x) + g(x)d$$

where

$$\begin{aligned}
h_1(x) &= \frac{1}{LC} V_C + (\frac{1}{RC} - a_0)[\frac{1}{C} i_L - \frac{1}{RC} V_C - \frac{1}{C} i_{LA}] \\
&\quad + \frac{1}{C} \dot{i_{LA}} \\
g_1(x) &= -\frac{V_{in}}{LC}
\end{aligned}$$

Clearly $g(x)$ is negative, but the sign of $h(x)$ is state dependent. This motivates an equivalent control u_{eq} policy that cancels out some terms to stabilize dynamics. u_{eq} can be found by setting $\dot{S}_1$ to 0.

$$\begin{aligned}
\dot{S_1} &= h_1(x) + g_1(x) u_{eq} = 0 \\
u_{eq} &= (\frac{1}{LC} V_C + (\frac{1}{RC} - a_0)[\frac{1}{C} i_L - \frac{1}{RC} V_C - \frac{1}{C} i_{LA}] \\
&\quad + \frac{1}{C} \dot{i_{LA}}) \times (-\frac{LC}{V_{in}})
\end{aligned}$$

Thus the control d is given by:

$$d = u_{eq} - K_1 sgn(S_1)$$

B. Switching Function S_2

The H-bridge requires only a first-order sliding surface because the motor current equation contains the control input u_2.

The sliding surface S_2 and its derivative are given below:

$$\begin{aligned}
S_2 &= i_{targ} - i_{LA} = e_i \\
\dot{S_2} &= -\dot{i_{LA}} = \frac{1}{L_A} V_C + \frac{R_A}{L_A} i_{LA} + \frac{K_e}{L_A} \omega - \frac{2V_C}{L_A} u_2
\end{aligned}$$

Consolidating terms:

$$\begin{aligned}
h_2(x) &= \frac{1}{L_A} V_C + \frac{R_A}{L_A} i_{LA} + \frac{K_e}{L_A} \omega \\
g_2(x) &= -\frac{2V_C}{L_A}
\end{aligned}$$

Once again, an equivalent control u_{eq} needs to be defined:

$$\dot{S_2} = h_2(x) + g_2(x) u_{eq} = 0$$
$$u_{eq} = 0.5 + \frac{R_A i_{LA} + \frac{K_e}{L_A} \omega}{2V_C}$$

Thus the control u_2 which outputs a 1 or 0 is given by:

$$u_2 = \{ \begin{array}{ll} 1, & \text{for } u_{eq} - K_2 sgn(S_2) > 0 \\ 0, & \text{for } u_{eq} - K_2 sgn(S_2) < 0 \end{array}$$

C. Tuning

These SMC controllers can both be tuned by selecting a_0. K can also be selected as long as the gain is not in a region that causes instability. Higher gains mean faster convergence but increased chatter, while lower gains achieve the opposite.

VII. Model Predictive Control Algorithm

As can be seen in **Algorithm 1**, the MPC is run with a LQR-type quadratic cost. This is to make sure that the cost function can be minimized with iterative methods. Of course, the quadratic cost function can morph under the provided constraints on all of the parameters so global optimality is not guaranteed. The chosen optimizer was IPOPT which is a popular nonlinear optimizer written in C++.

The controller can be tuned with various tradeoffs. Increasing the horizon and decreasing the time step increases the optimizer's performanc with slower dynamics and aids in stability and convergence. The controller will behave more like a closed loop controller. The cost Q penalizes not hitting the desired angular velocity while the cost R_i penalizes extreme currents and the cost R_v penalizes extreme voltages.

VIII. Results and Discussion

Figure 4 shows the results of the Model Predictive and Sliding Mode Control (MP-SMC) controller running on a simulated plant. State feedback was given to the controller at each time step. The sliding mode controllers seem to work very well, converging to the voltage and current commands very quickly. The model predictive portion of the controller unfortunately does have some offset error and this is believed to be because the terminal cost is equal to the voltage and

Algorithm 1 Cascade Outer-Loop: Model Predictive Control

Input: State $\mathbf{x} = [\omega, \theta]^T$, Target ω_{target}
Output: Optimal target i^*_{ref} and $v^*_{c,ref}$
Params: Horizon N, Time step ΔT, Weights Q, R_i, R_v
Initialize optimization problem $\mathcal{P}$, Cost $J \leftarrow 0$
Define state trajectory $\mathbf{X} = \{\mathbf{x}_0, \ldots, \mathbf{x}_N\}$
Define control trajectory $\mathbf{U} = \{\mathbf{u}_0, \ldots, \mathbf{u}_{N-1}\}$
Set initial condition constraint: $\mathbf{x}_0 = \mathbf{x}$
for $k = 0, 1, \ldots, N-1$ **do**
    $\omega_k \leftarrow \mathbf{x}_k[0], \quad i_{ref,k} \leftarrow \mathbf{u}_k[0], \quad v_{c,ref,k} \leftarrow \mathbf{u}_k[1]$
    % System Dynamics Constraint (RK4)
    $\mathbf{x}_{k+1} = F_{discrete}(\mathbf{x}_k, \mathbf{u}_k, \Delta T)$
    % Inner-Loop SMC Reachability Constraints
    $v_{c,ref,k} \geq (R_a i_{ref,k} + K_e \omega_k) + \epsilon$
    $v_{c,ref,k} \geq -(R_a i_{ref,k} + K_e \omega_k) + \epsilon$
    % Physical Hardware Bounds
    $-I_{max} \leq i_{ref,k} \leq I_{max}$
    $0 \leq v_{c,ref,k} \leq V_{in}$
    % Accumulate Stage Cost
    $J \leftarrow J + Q(\omega_k - \omega_{target})^2 + R_i(i_{ref,k})^2 + R_v(v_{c,ref,k})^2$
end for
% Add Terminal Cost
$J \leftarrow J + Q(\mathbf{x}_N[0] - \omega_{target})^2$
Solve $\min_{\mathbf{X},\mathbf{U}} J$ subject to all constraints
if Solver converges optimally **then**
    $i^*_{ref} \leftarrow \mathbf{u}_0[0]$
    $v^*_{c,ref} \leftarrow \mathbf{u}_0[1]$
    Warm-start next solver iteration with $\mathbf{X}^*, \mathbf{U}^*$
else
    % Fallback ride-through mechanism
    Keep previous targets $i^*_{ref}, v^*_{c,ref}$
end if
return $i^*_{ref}, v^*_{c,ref}$

current effort cost such that any amount of effort would increase the cost more than the terminal cost decreases. Varying the cost parameters does not completely solve this problem however, so future work developing better cost functions is necessary.

[(Figure 6)] : This controller uses an expression for the voltage setpoint similar to the reachability constraint of mpc where $V_c > i_{La}R_a + K_e\omega_{set}$ must be enforced. For controlling the switching events of the fullbridge it checks if $\omega > \omega_{set}$ and switches the full bridge polarity if true.

IX. Next Steps

For the Mrover team, the next steps would be to implement a hardware prototype of this system and work on incorporating higher levels of fidelity into their modeling.

References

[1] D. Montoya-Acevedo, W. Gil-González, O. D. Montoya, C. Restrepo and C. González-Castaño, "Adaptive Speed Control for a DC Motor Using DC/DC Converters: An Inverse Optimal Control Approach," in

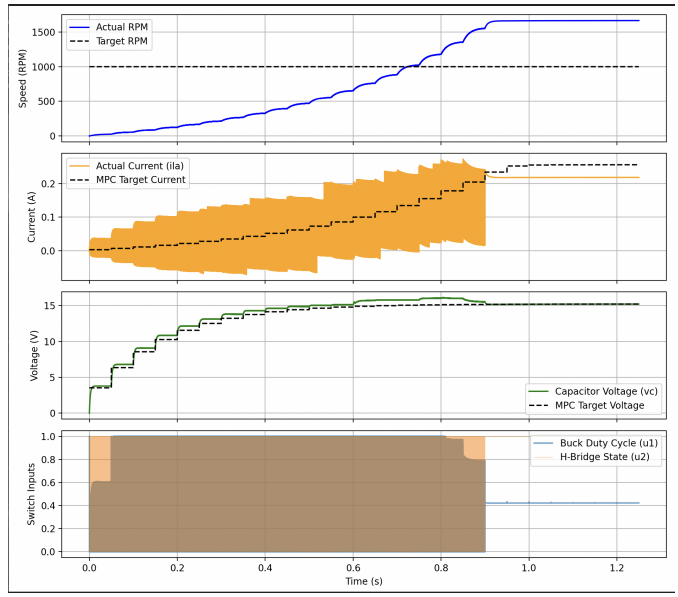


Fig. 5. Graphs showing (1) angular velocity tracking (2) current tracking (3) voltage tracking (4) switching tracking

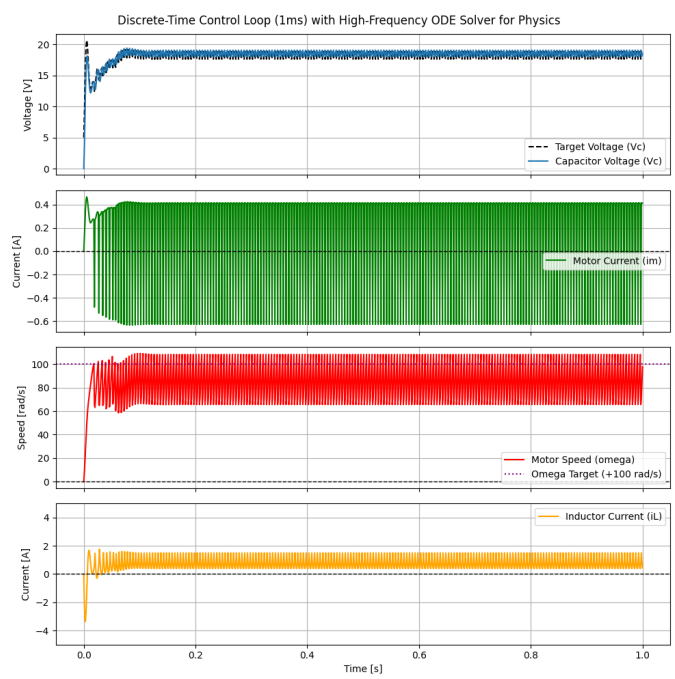


Fig. 6. PI Voltage control with Proportional Full bridge control

IEEE Access, vol. 12, pp. 154503-154513, 2024, doi: 10.1109/ACCESS.2024.3482982.

[2] D. Montoya-Acevedo, W. Gil-González, O. D. Montoya, C. Restrepo and C. González-Castaño, "Adaptive Speed Control for a DC Motor Using DC/DC Converters: An Inverse Optimal Control Approach," in IEEE Access, vol. 12, pp. 154503-154513, 2024, doi: 10.1109/ACCESS.2024.3482982.

[3] K. M. Smedley and S. Cuk, "One-cycle control of switching converters," in IEEE Transactions on Power Electronics, vol. 10, no. 6, pp. 625-633, Nov. 1995, doi: 10.1109/63.471281.